



Boolean Algebra



Brief History of Boolean Algebra

- According to Huntington, the term “Boolean Algebra” was first suggested by Sheffer in 1913. Boolean Algebra has been important in the growth of digital electronics in designing digital circuit, and is provided for in all modern programming languages.
- A set of rules or Laws of Boolean Algebra Law expressions have been invented to help reduce the number of logic gates.

Boolean Properties	ANDed form	ORed form
Identity	$A=A$ $A'=A'$	
Commutative	$AB = BA$	$A+B = B+A$
Associative	$A(BC) = B(AC) = C(AB)$	$A+(B+C) = B+(A+C) =$ $C+(A+B)$
Idempotent	$AA = A$	$A+A = A$
Double Negative	$A'' = A$	
Complementary	$AA' = 0$	$A+A' = 1$
Law of Intersection	$A1 = A$ $A0 = 0$	

Union

$$A+1 = 1$$
$$A+0 = A$$

Distributive

$$A(B+C) = (AB)+(AC)$$

$$A+(BC) = (A+B)(A+C)$$

Absorption

$$A(A+B) = A$$

$$A+(AB) = A$$

Common Identities

$$A(A'+B) = AB$$

$$A+(A'B) = A+B$$

De Morgan's
Theorem

$$(AB)' = A'+B'$$

$$(A+B)' = A'B'$$

Identity Law

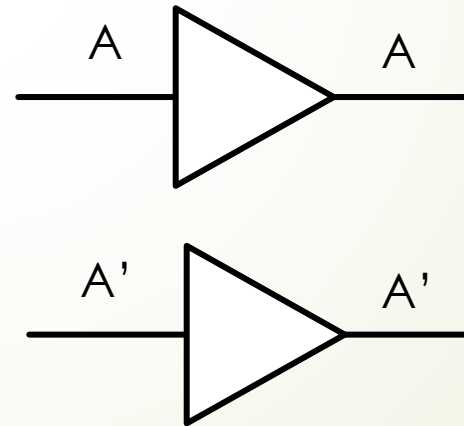
- Law of Identity states that each thing is identical with self

Expression

$$A = A$$

$$A' = A'$$

Logic Circuit Representation



Commutative Law

- Commutative Law allowing a change in position for addition and multiplication.

Expression:

$AB = BA$ - The order in which two variables are AND'ed makes no difference

$A+B = B+A$ - The order in which two variables are OR'ed makes no difference

Associative Law

- This law allows the removal of brackets from an expression and regrouping of the variables.

Expression:

$$A + (B + C) = (A + B) + C = A + B + C \quad (\text{OR Associate Law})$$

$$A(B.C) = (A.B)C = A . B . C \quad (\text{AND Associate Law})$$

Idempotent Law

- An input that is AND'ed or OR'ed with itself is equal to that input

Expression:

$A + A = A$ - A variable OR'ed with itself is always equal to the variable

$A . A = A$ - A variable AND'ed with itself is always equal to the variable

Double Negative or Double Negation

- A term that is inverted twice is equal to the original term

Expression:

$A'' = A$ - A double complement of a variable is always equal to the variable

Complementary Law

- A term AND'ed with its complement equals "0" and a term OR'ed with its complement equals "1"

Expression:

$AA' = 0$ - A variable AND'ed with its complement is always equal to 0

$A + A' = 1$ - A variable OR'ed with its complement is always equal to 1

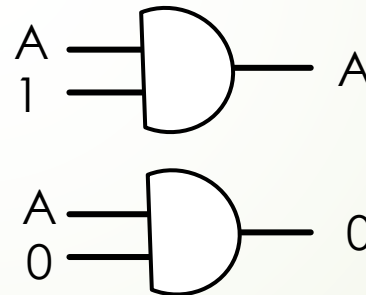
Law of Intersection

- A rule in logic that the value of one input is 'AND' to '1' the result will be equal to its value 'OR' to '0' the result is equal to '0'.

Expression:

$$A1 = A$$

$$A0 = 0$$



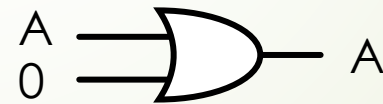
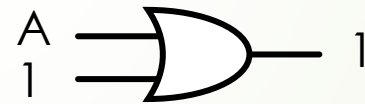
Union

- A rule in logic that the value 'OR' to logic '1' the result value is equal to '1' and if the value 'OR' to logic '0' the result value is equal to its value.

Expression:

$$A+1 = 1$$

$$A+0 = A$$



Distributive

- This law permits the multiplying or factoring out of an expression.

Expression:

$$A(B + C) = A.B + A.C \quad (\text{OR Distributive Law})$$

$$A + (B.C) = (A + B).(A + C) \quad (\text{AND Distributive Law})$$

Absorption

- This law enables a reduction in a complicated expression to a simpler one by absorbing like terms.

Expression:

$$A + (A.B) = A \quad (\text{OR Absorption Law})$$

$$A(A + B) = A \quad (\text{AND Absorption Law})$$

Common Identities

- A rule where A ANDed to its Complement and ORed to the B input it gives a result of A ORed B or A ANDed B.

Expression:

$$A(A' + B) = AB$$

$$A + (A'B) = A + B$$

De Morgan's Theorem

- ▶ Two separate terms NOR'ed together is the same as the two terms inverted (Complement) and AND'ed for example: **$A' + B' = AB'$**
- ▶ Two separate terms NAND'ed together is the same as the two terms inverted (Complement) and OR'ed for example: **$AB' = A' + B'$**

Boolean Algebra Simplification

Examples:

$$X = (AB) + (AC)$$

$$(A+A)(A+C)(B+A)(B+C)$$

$$A(A+C)(B+A)(B+C)$$

$$A(B+A)(B+C)$$

$$X = A(B+C)$$

Distributive law (ORed form)

Idempotent Law: $A+A = A$

Absorption Law

Absorption Law

Simplified Expression

Example 2:

$$X = (A'B) + (BC') + (BC)$$

$$(A'B) + (B+B)(B+C)(C'+B)(C'+C)$$

$$(A'B) + (B+B)(B+C)(C'+B)1$$

$$(A'B) + B(B+C)(C'+B)1$$

$$B(B+C)(C'+B)1$$

$$B(C'+B)1$$

$$B1$$

$$X = B$$

Distributive Law (ORed form)

Complementary Law: $C' + C = 1$

Idempotent Law: $B + B = B$

Absorption Law

Absorption Law

Absorption Law

Identity Law: $B1 = B$

Simplified Expression

Example 3: De Morgan's Theorem

$$\begin{aligned} X &= ((A'+B)'A)' \\ &\quad ((A'+B)''+A')'' \\ &\quad A'+B+A' \\ X &= A'+B \end{aligned}$$

De Morgan's Theorem: Convert AND to Or
Double Negation

Idempotent Law : $A'+A' = A'$

Simplified Form